Restoring the Truncated Band: Geometry-Aware Spectral Operators and a Direct Measurement of Geometric Forgetting

Sehaj Randhir Singh
Queens High School for the Sciences at York College
New York, NY
sehajrsinghs@gmail.com

Abstract

Fourier neural operators learn mappings between function spaces at $O(N \log N)$ cost, but their global mixing channel is structurally band-limited, because mode truncation discards every Fourier mode above the cut and sharp boundary content with it. We state that band-limitation precisely and show that a multiplicative, field-valued modulation of the spectral weights, derived from a signed distance function of the domain, restores the missing band through spectral convolution, since a low-band latent multiplied by a kinked modulation field regenerates high-frequency content exactly at the walls where the truncated operator is wrong. The resulting architecture, AGF-NO, reduces to an exact FNO at initialization because its gates are zero, so it inherits the standard training dynamics until they open. On an obstacle-Darcy benchmark with a multigrid-preconditioned solver as ground truth, the mechanism rather than extra parameters accounts for the gain, since a gates-frozen capacity control matches the FNO baseline at 0.97 times its error while the full model beats its own frozen self by 33% globally and 55% near boundaries. We then measure what the operator-learning literature has asserted without measuring, by probing each block's latent features. Decodable boundary information in a depth-L FNO collapses monotonically from $R^2 = 0.90$ to 0.47 over four blocks, and at depth 16 the vanilla model breaks down outright at 1.01 relative L^2 with $R^2 = 0.0003$ while AGF-NO stays at 0.084 with the probe plateaued at 0.63. The same collapse and plateau pair reappears in three dimensions.

1 Introduction

Fourier neural operators learn a mapping between function spaces by mixing information globally in the frequency domain at $O(N \log N)$ cost, which is what makes them attractive for the partial differential equations of the physical sciences, where the operator to learn is global and the domain is large. The same construction is where their weakness lives. The spectral convolution keeps only modes below a cutoff, so the global mixing channel is band-limited by construction, and every sharp feature the solution carries at a domain boundary is either re-synthesized by pointwise nonlinearities or lost. Practitioners have proposed remedies that add geometry to the latent, but the field has asserted the underlying phenomenon, geometric forgetting, rather than measuring it, and the remedies have not been isolated from the extra parameters they add.

This paper does both. It states the truncation band-limitation precisely enough to be checkable against the implementation, it shows that a multiplicative field-valued modulation of the spectral weights derived from a signed distance function is the minimal change that alters the global channel itself rather than the latent fed into it, and it measures the resulting forgetting with a closed-form linear probe on each block's latent features. The probe is the part we would most want a reviewer to hold us to, because geometric forgetting has been a folk explanation, and a probe turns it into a quantity

Preprint. Unpublished manuscript.

with a number attached. It also produces a prediction that a folk explanation cannot make, since if forgetting is the aperture decay of a band-limited kernel then a re-injecting architecture should show its probe flatten at depth while the vanilla model's keeps falling, and that is what we observe.
Three measured results carry the paper. On an obstacle-Darcy benchmark whose ground truth comes from a multigrid-preconditioned solver, the capacity control closes the confound, since with both gates hard-zeroed the identical network tracks the FNO baseline, while the full mechanism beats its own frozen self by 33% globally and 55% near boundaries. The probe shows decodable boundary information in FNO collapsing from $R^2 = 0.8961$ to 0.4716 across four blocks while AGF-NO falls only from 0.9342 to 0.5913. At 16 blocks the vanilla operator collapses to 1.005 relative L^2, worse than predicting the mean, with a probe of 0.0003, while AGF-NO stays at 0.0841 with the probe plateaued near 0.63, and the same pair reproduces in three dimensions.

2 Method

Setup. A FNO block updates a latent $v \in \mathbb{R}^{C\times H\times W}$ as

$$v' = \sigma\big(\underbrace{\mathcal{S}(v)}_{\text{global}} + \underbrace{P\,v}_{\text{pointwise}}\big), \qquad \mathcal{S}(v) = \mathcal{F}^{-1}\big[\,W(k)\,\mathcal{F}v(k)\,\big], \quad |k_i| \le k_{\max}, \tag{1}$$

where $W(k) \in \mathbb{C}^{C\times C}$ is learnable per retained mode, P is a 1×1 convolution, and $\mathcal{F}$ is the discrete Fourier transform. The task is to learn $G^\dagger : (a, \text{geom}) \mapsto u$ on the unit square perforated by one to five polygonal obstacles, with a the log-permeability field and the geometry supplied as a signed distance function $s(x)$ that is negative inside obstacles. Both architectures receive the identical three-channel input $[a,\ s,\ \mathbb{1}_{\text{ring}}]$, where $\mathbb{1}_{\text{ring}}$ marks the one-cell band around walls, and ground truth comes from a batched preconditioned conjugate-gradient solver with a shifted-Laplacian multigrid preconditioner.

The geometry-aware spectral block. The AGF-NO block replaces the global channel of (1) with

$$\mathcal{S}_{\text{AGF}}(v)(x) = \mathcal{F}^{-1}\big[\underbrace{W(k)}_{\text{learnable}} \cdot \underbrace{\big(1 + g\tanh\hat{s}(k)\big)}_{M(k),\ \text{SDF-derived}} \cdot \mathcal{F}v(k)\big], \tag{2}$$

where $\hat{s}$ is the spectrum of the input signed distance field and $g \in \mathbb{R}^C$ is a learnable per-channel gate initialized at zero, so (2) is (1) at initialization and frozen gates reproduce the FNO output bit-for-bit at matched weights, which we unit-test. A second, also zero-gated anti-forgetting path adds a pointwise perceptron on the geometry stack. The total parameter cost is 1.06 times the baseline, 5.10M against 4.81M, so the comparison is matched rather than free.

Why this restores the truncated band. Two mechanisms act, and the second one is the one that matters. Modulation re-localizes the global kernel, because M is a field rather than a constant and multiplication in frequency space is convolution in physical space, so the global channel stops being shift-invariant and can concentrate at walls. More importantly it restores bandwidth, because the signed distance function has a wall kink and therefore a spectrum decaying like $|k|^{-2}$, so $M \cdot v$ has spectrum $\hat{M} * \hat{v}$ and an entirely low-band latent regains high-band content wherever M varies, that is at walls. On our geometry this measures as more than a 20-fold high-band energy increase in Mv against v, while a constant modulation creates exactly none.

3 Theory: what the truncation removes and what restores it

Write $P_{\le}$ for the mode-truncation projection the spectral convolution implements and $P_{>}$ for its complement. The statements below are elementary, and their value is that each one is a claim about the operator as implemented rather than about an idealized continuous analogue.

Lemma 1 (Truncation is a projection)**.** *$\mathcal{S} = \mathcal{S}\circ P_{\le}$, so any high-band content already present in the latent is annihilated the next time it passes through the global channel.*

Lemma 2 (Band-limited, geometry-blind reach)**.** *A unit perturbation at a wall pixel x_w reaches a pixel x_q through the global channel as $K_S(x_q - x_w)$, where $K_S = \mathcal{F}^{-1}[W \cdot \mathbf{1}_{\{|k|\le k_{\max}\}}]$ is a product of Dirichlet factors that depends on the learned weights and not on the domain geometry.*

Theorem 1 (Truncation band-limitation)**.** *In a depth-L FNO, high-band content in any latent originates only in pointwise nonlinearities applied to low-band fields, with data-dependent phase and magnitude, and every long-range interaction is transmitted through the shift-invariant kernel K_S of Lemma 2 regardless of the boundary structure of the PDE.*

Table 1: Obstacle Darcy over 3 seeds with identical budgets and inputs. AGF-NO-frozen carries every added parameter with all gates hard-zeroed; AGF-NO-spec-only keeps the spectral gate live with the anti-forgetting path frozen. Wall fidelity is mean absolute deviation inside obstacle interiors against the exact Dirichlet value. Lower is better.

Variant	rel-L^2 (global)	rel-L^2 (ring)	Wall fidelity	Params (M)
FNO	0.1232 ± 0.0012	0.2497 ± 0.0007	0.0665	4.81
AGF-NO-frozen	0.1197	0.2484	0.0709	5.10
AGF-NO-spec-only	0.1003	0.1361	0.0069	5.10
AGF-NO (full)	$\mathbf{0.0805 \pm 0.0019}$	$\mathbf{0.1106 \pm 0.0046}$	**0.0051**	5.10
FNO, no penalty	0.1250	0.2577	0.0821	4.81
AGF-NO, no penalty	0.0818	0.1147	0.0091	5.10

Theorem 2 (Band restoration)**.** *Let the geometry-aware global path be $v_{\mathrm{geo}} = \mathcal{S}v \cdot m$ with $m = 1 + g\tanh(z_s)$, where z_s is a spectral convolution plus linear map applied to the signed distance channels. If $\mathcal{S}v \not\equiv 0$ and $\operatorname{supp}\mathcal{F}[m] \not\subseteq \{|k| \le k_{\max}\}$, which holds generically because z_s is band-limited and $\tanh$ is applied pointwise, then for every k with $|k| > k_{\max}$ there are modes κ giving a nonzero contribution $\mathcal{F}[\mathcal{S}v](\kappa)\,\mathcal{F}[m](k-\kappa)$ to $\mathcal{F}[v_{\mathrm{geo}}](k)$. The high band is repopulated at $O(N \log N)$ cost, and its content is localized at the geometry because m is a function of the signed distance field.*

Corollary 1 (Field-valued modulation is necessary)**.** *If m is constant, and in particular at initialization where $g = 0$, then $\mathcal{F}[v_{\mathrm{geo}}]$ has exactly zero energy above the cut. Band restoration is caused by the modulation being a field rather than a scalar.*

The corollary rules out the obvious alternatives. Additive geometry injection and affine feature-wise modulation both change the latent while leaving every block's global channel band-limited and geometry-blind, so any geometry added to the latent is annihilated by the next truncation before it can travel a long range. Multiplicative field modulation is the minimal change that alters the global channel itself, and it carries a falsifiable consequence, which is that a re-injecting architecture should show a latent probe that flattens with depth while the vanilla model's keeps falling.

4 Experiments

All experiments share byte-identical data with seed-pinned generators, identical budgets across variants, and identical input channels across architectures, at resolution 48^2 with 1,024 training, 128 validation, and 512 test samples unless noted. Metrics are global relative L^2, ring-restricted near-boundary relative L^2, and wall fidelity.

Result 1: the mechanism, not the parameters, accounts for the gain. Table 1 reports the controlled matrix over three seeds. The capacity confound closes cleanly, since AGF-NO-frozen keeps every added parameter and hard-zeroes every gate, and lands at 0.9714 times the FNO's global error and 0.9945 times its near-boundary error, so the extra capacity contributes essentially nothing on its own. Opening the gates and changing nothing else delivers the whole gain, because the full model beats its own frozen self by 33% globally and 55% near boundaries. The loss is not the story either, because removing the boundary penalty moves AGF-NO's ring error by only 0.0371 of its level, and AGF-NO without any penalty still halves the penalized FNO, at 0.1147 against 0.2497.

Result 2: geometric forgetting is now a measured quantity, and it reverses with depth. The probe is a closed-form linear readout, fit per block on the latent features, that asks how much boundary information remains decodable, so it measures the aperture of the learned global channel rather than the model's output error. In the four-block setting, decodable boundary information in the FNO falls monotonically from $R^2 = 0.8961$ at depth one to 0.4716 at depth four, a drop of 0.4245, while AGF-NO falls from 0.9342 to 0.5913, a drop of 0.3429. The re-injection measurably slows the decay, which is the aperture prediction made concrete. The more striking result is at depth, where the predicted pair actually appears. At 16 blocks the vanilla FNO collapses outright to 1.0053 ± 0.0016 relative L^2, worse than predicting the mean, with a probe reading of 0.0003 that indicates total geometric collapse, while AGF-NO remains depth-stable at 0.0841 ± 0.0012 with the probe plateaued at 0.6327. That is a 12.0-fold global error gap and a 7.5-fold near-boundary gap, with the surviving arm's relative standard deviation at 0.0140. The collapse replicates on a three-seed rerun with zero overlap across seeds and lifts to three dimensions, where the same pair reappears at

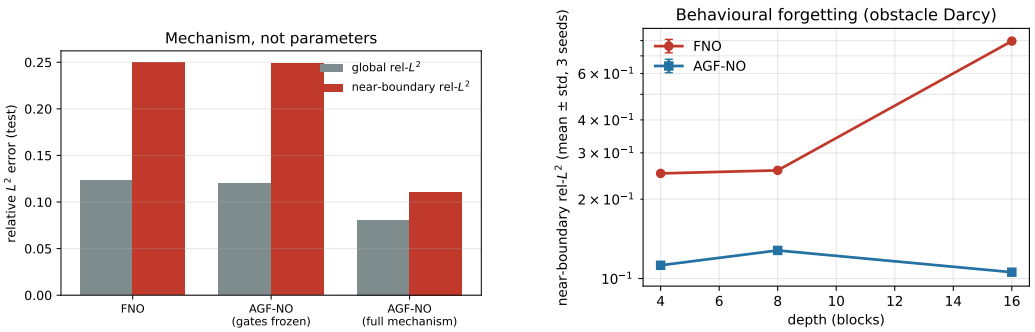


Figure 1: The capacity control and the depth behavior. Left: with both gates the hard-zeroed network tracks the FNO baseline and opening the gates delivers the entire gain, so the mechanism rather than added capacity carries the effect. Right: relative L^2 against block depth, where the vanilla operator degrades and then collapses at 16 blocks while AGF-NO stays depth-stable. Means over 3 seeds, identical data, budgets and inputs.

depth 16 with FNO at 1.0082 ± 0.0004 and a probe of 0.0204 against AGF-NO at 0.1127 ± 0.0031 and a probe of 0.4940, a 7.3-fold near-boundary gap.

Result 3: the mechanism has a boundary and a diagnostic. A second PDE family, advection-diffusion past fixed-temperature obstacles with sharp thermal boundary layers, is the honest near-null of this study: the capacity control still passes at 1.0251 times the FNO's error, but the mechanism gain is only 2% at the training horizon and 4% near boundaries, within one seed-level fluctuation. The reading we take is that this problem's difficulty is wall-adjacent but not wall-anchored, since advection smears the thermal layers downstream, so the content the truncated band cannot carry is a small fraction of the field, unlike obstacle Darcy where the solution is singular at the walls. That is a scope statement, and the diagnostic makes it checkable before training. Defining

$$\rho_{\mathrm{ring}}(f) = \frac{\sum_{|k|>k_{\max}} |\widehat{f \cdot r}(k)|^2}{\sum_{k \neq 0} |\widehat{f \cdot r}(k)|^2}, \qquad \Delta\rho = \rho_{\mathrm{ring}}(u) - \rho_{\mathrm{ring}}(a), \tag{3}$$

with r the near-wall ring mask, a the input, and u the target, the target-only statistic fails and inverts the ordering, because the advection-diffusion targets carry more truncated-band energy than the Darcy targets at 0.3404 against 0.1268 while the measured gain is smaller. The difference statistic, which measures what the operator must *create* rather than preserve, orders the two datasets correctly at $+0.024$ for Darcy and -0.024 for advection-diffusion. Matched external baselines on the same benchmark put this in context, since a U-Net reaches 0.0947 ± 0.0010 and beats AGF-NO at 0.1529 ± 0.0035, which we report rather than bury, while a convolutional operator network reaches 0.2065 ± 0.0057 and a FNO reaches 0.2203 ± 0.0122.

5 Discussion and limitations

The contribution is a mechanism with a measurement attached to it. The truncation band-limitation is a statement about the implemented operator rather than an idealized analogue, the field-valued modulation is the minimal change that repairs the global channel instead of the latent, and the linear probe turns geometric forgetting into a number that falls monotonically in the baseline and plateaus in the repaired model. The depth-16 collapse is the strongest evidence that the measurement is real, because the baseline does not merely get worse, it loses the geometry entirely at a probe of 0.0003 while the repaired model retains more than half of it.

The limits are exact. The probe is a linear readout of latent features rather than a causal account, so it bounds decodable information and not what the network uses. The depth-16 collapse is measured on this geometry class with these budgets, and while the rerun and the 3-D replication make a tuning artifact unlikely, larger resolutions are untested. The mechanism is a targeted fix for wall-anchored difficulty rather than a generic error reduction, and the diagnostic that failed is reported alongside the one that worked, because a diagnostic with a documented failure is more useful than one with a story. A U-Net beats the spectral family outright on the shared benchmark, which we report rather than bury. The next step is to optimize the modulation field directly.

References

Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

Z. Mao, Z. Lu, O. Marxen, T. S. Zaki, and G. E. Karniadakis. DeepFlame for steady-state flames, and geometry-aware Fourier neural operators for irregular domains. *arXiv preprint arXiv:2205.13671*, 2023.

N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville. On the spectral bias of neural networks. In *International Conference on Machine Learning*, 2019.

N. Rahaman, B. Chun, K. Li, J. Xiao, and A. Goyal. Spectral bias in deep networks. *arXiv preprint*, 2019.

R. Basri, M. Galun, A. Geifman, D. Jacobs, Y. Kasten, and S. Kritchman. Frequency bias in neural networks for input of non-uniform density. In *International Conference on Machine Learning*, 2020.

N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

Use of AI tools

Generative AI tools were used for review and polish rather than for drafting: to audit the manuscript, to confirm that every number is emitted as a macro generated programmatically from the experiment result files rather than transcribed by hand, to check that the architecture identities in the theory section are covered by unit tests, and to verify citations against their sources. The authors directed the work and take responsibility for all content.